

University of Sheffield

Multisensors Assignment



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1 Task 1: Kalman Filter Design

1.1 Initial Linearization

To design an an integrated GPS/IMU/Air-data navigation system let's first consider the system dynamics in the following format:

$$\dot{x} = f(x(t), c(t), t) + g(x(t), c(t), \omega(t)) \quad (1)$$

$$d(t) = h(x(t), t) \quad (2)$$

$$d_m(t_k) = d(t_k) + \nu(t_k), k = 1, 2, \dots \quad (3)$$

where 1 is the kinematic equation, 2 is the observation equation and 3 is the measurement model. Here, x is the state variables, c is the input variables, d the output variables, d_m the measured output, ω is the system noise and ν the output measurement noise. To use the Kalman Filter, the first step involves the derivation of the Jacobian Matrices for the nonlinear system. Using equations 1 and 2 to calculate f and g and using 3 for calculating h , three variables are calculated, namely $F = \frac{\partial f}{\partial x}$, $G = \frac{\partial g}{\partial \omega}$ and $H = \frac{\partial h}{\partial x}$. These three matrices mainly consist of equations which are nothing but the elements of the matrices. A few of the elements in these matrices are as follows:

- The element in 5th row, 4th column of F matrix is $-r_m = -(r + \omega_r)$
- The element in 6th row, 7th column of F matrix is $-g * \cos(\theta) * \sin(\phi)$
- The element in 4th row, 6th column of G matrix is $-v$
- The element in 9th row, 5th column of G matrix is $-\frac{\sin \phi}{\cos \theta}$
- The element in 7th row, 7th column of H matrix is 1
- The element in 11th row, 4th column of H matrix is $-w/(u^2 * (w^2/u^2 + 1))$

where u , w are the ground speed earth components, ϕ and θ are the roll and pitch angle respectively, g is the gravitational force, r is the rotational yaw rate in rad/s and ω_r is the system noise for this yaw rate.

1.2 Kalman Filter

Before applying the Kalman Filter to the system, represented by the dataset dataTask1, a few necessary engineering decisions have to be made. There are a few assumptions that need to be made during the design of the Kalman Filter. The state variables that could be directly measured, i.e. x , y , z , ϕ , θ and ψ were the same as the noise measurements during the first time instance. The states u , v , and w were initially estimated to be V_{TAS} (Air Speed) = 95m/s, 0 and 0 respectively, and the wind speeds that were also assumed to be zero along with the gravitational force of $9.81m^2/s$.

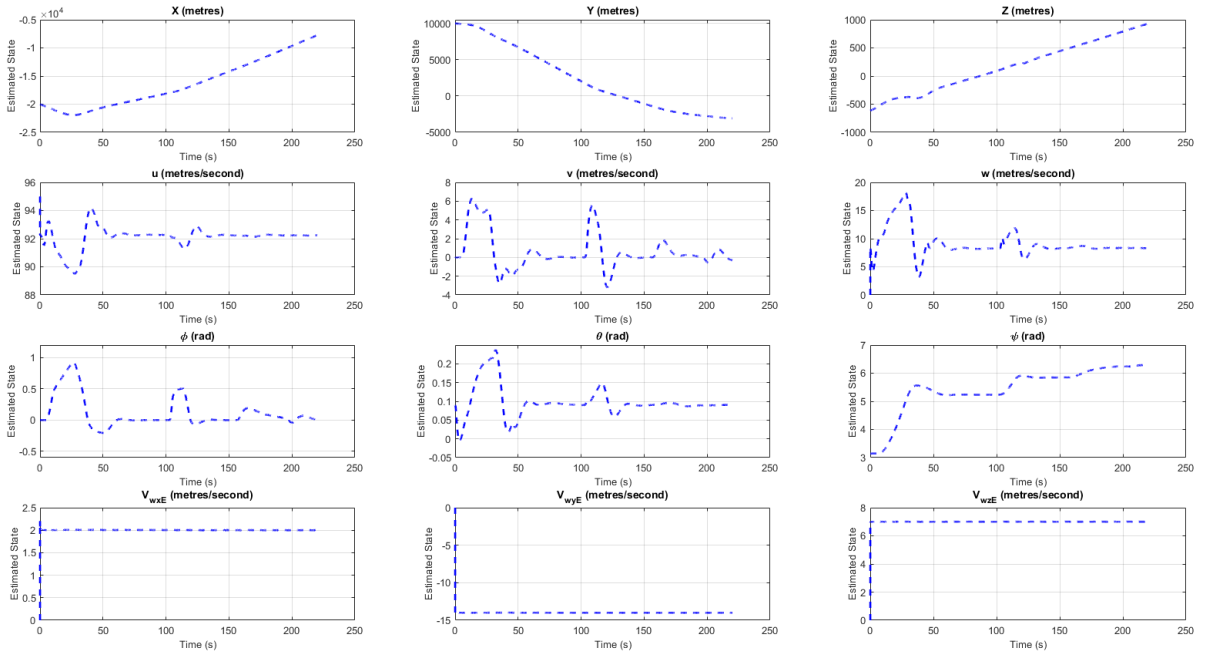


Figure 1: State Estimates after Kalman Filter Design

By using this engineering estimation, standard deviations of either low or high values were made based. For this case, since the values of x , y , z , ϕ , θ , and ψ were known, the standard deviations relating to them were considered to be low (in this case 1), whereas the remaining 6 states were assumptions and hence their standard deviation values were considered to be large (in this case 100).

Considering these estimates and assumptions, the resultant states and their estimates for a particular time frame were generated by applying the recursive Kalman Filter process to the system, as depicted in Figure 1. From the figure, after carefully observing the plots a few conclusions can be made. The aircraft position changes concerning time. The x position and z position increase with respect to the original position as time passes but

then the y position decreases with time. The estimation of V_{TAS} which was taken to be $95m^2/s$ seems to be almost correct as the plotting of state u, which was the ground speed earth component, stabilizes to a state of $92m^2/s$ and the assumptions of v and w ground speed earth components were correctly assumed to be 0. The roll and pitch angle were also correctly assumed to be 0 and 0.1 whereas the yaw angle is corrected to 6.3 due to Kalman Filter. The Original assumptions of 0 taken for the wind speeds turned out to be wrong and the Kalman Filter corrected them to the right values.

2 Task 2: Removal of Sensor Biases

2.1 Normal and Biased sensor Comparison

In Task 1, the Kalman Filter was used to handle the noisy measurements. It is known that the sensors generally used are gyroscopes which are prone to face issues like drifts, sensor aging, and many other such problems causing biases in the values that are read from the sensor. To handle measurements having sensor biases let's first consider comparing two different datasets, one without bias as seen in Task1 to another having biases as given by the dataTask2 through the Kalman Filter. By running both the datasets through the Kalman Filter, some states have a huge difference between the values, some have a very small difference and some have no difference, as seen in Figure 2. The x, y, and z positions have no difference between the datasets, whereas there is a huge difference seen in the ground speed of u, v, and w shows a huge difference between the values in the datasets and a slight difference between the values in the roll, pitch and yaw angles. The wind speeds of the new biased data do not show a consistent value like that of the one seen in Task1 although they get corrected from the original estimation of 0. The highlighted plots in Figure 2 clearly show the states in which the difference between nominal and biased sensor has a maximum difference between the estimation values.

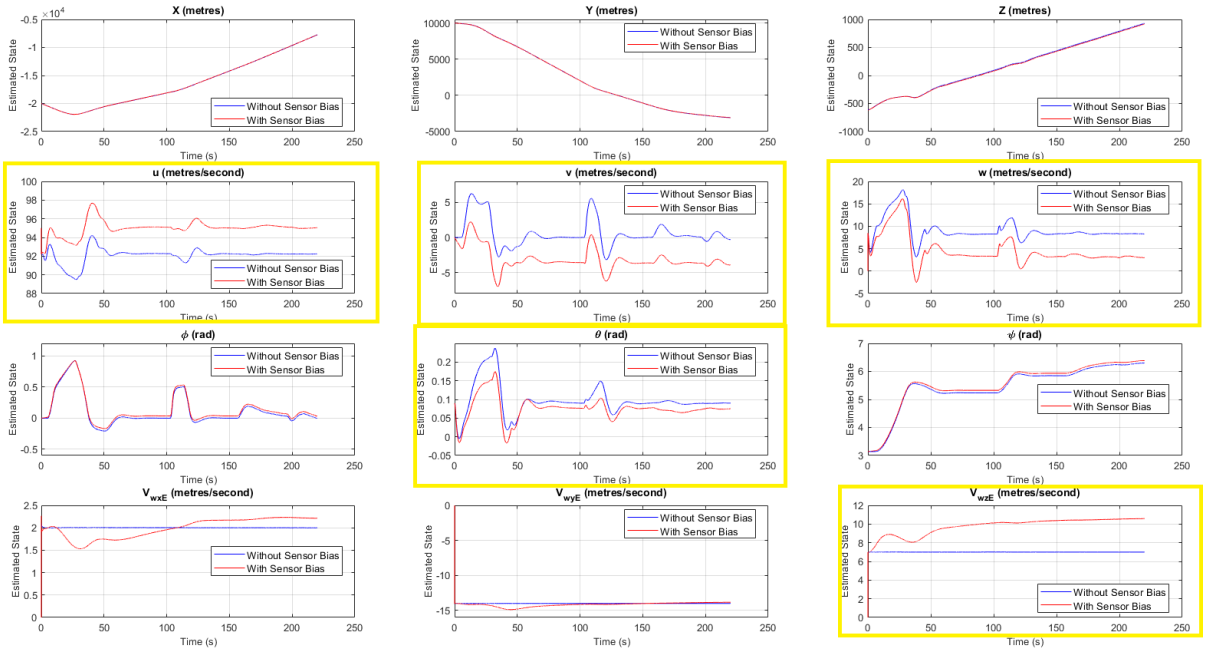


Figure 2: State Estimates after Kalman Filter Design without removal of Sensor Bias

The difference that arises occurs due to biases present in the sensors. By addition of 6 states to the system during the process of Kalman Filter, it is observed that Kalman Filter mitigates the effect of the biases during the calculation of the estimation of the states.

2.2 Biased Sensor Mitigation

To counter the effect of bias as previously seen, using these bias terms along with state estimation and also considering the rate of change of them to be 0, the following assumptions are made for the states and biases:

$$x = [x_E, y_E, z_E, u, v, w, \phi, \theta, \psi, b_{A_x}, b_{A_y}, b_{A_z}, b_p, b_q, b_r, V_{wxe}, V_{wye}, V_{wze}]$$

where the additional terms like b_{A_x} are the biases for the system, and:

$$[\dot{b}_{A_x}, \dot{b}_{A_y}, \dot{b}_{A_z}, \dot{b}_p, \dot{b}_q, \dot{b}_r] = [0, 0, 0, 0, 0, 0]$$

are the initial guesses that are taken for the biases during the process of applying the Kalman Filter Scheme.

Now, looking at how to Kalman Filter effects the biased terms is represented in Figure 3. It is seen from the figure that the 6 biases occurring in the inputs first show an initial spike which means that the Kalman Filter is trying to stabilize and mitigate the effect of biases which it eventually does after a very small period of time and brings the state of the biases to the required value. The plots in the figure have a shift of about 50 time instance ahead (around 5 seconds) to clearly indicate the initial spike in the beginning before settling down to a definitive value and since these biases eventually reach their potential value, it is hence proved that the biases have been finally mitigated.

To see how the mitigation is seen from the perspective of the states of the system, it is best to compare the current process with the situation in the previous two instances where one of the conditions was sensor reading without bias and the other with bias without it's mitigation. This comparison is represented in Figure 4.

As expected, it is seen that the result of mitigation of biases brings the value of the state estimates similar to that of the situation during Task1 after the application of the

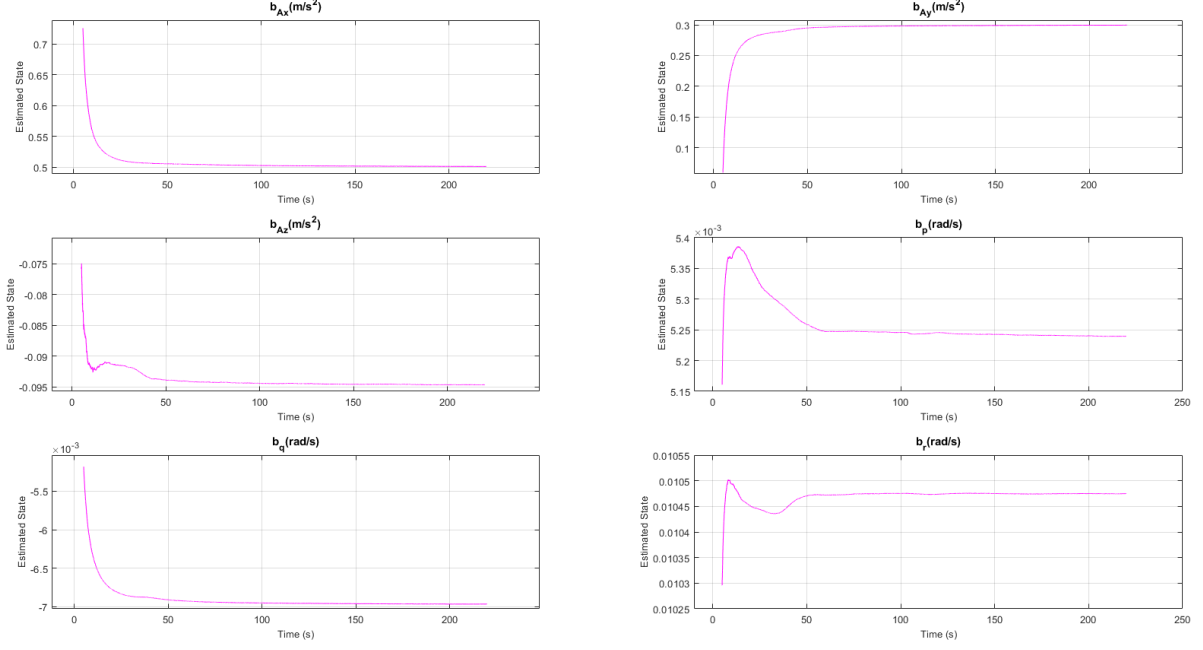


Figure 3: Bias Estimation after application through Kalman Filter scheme

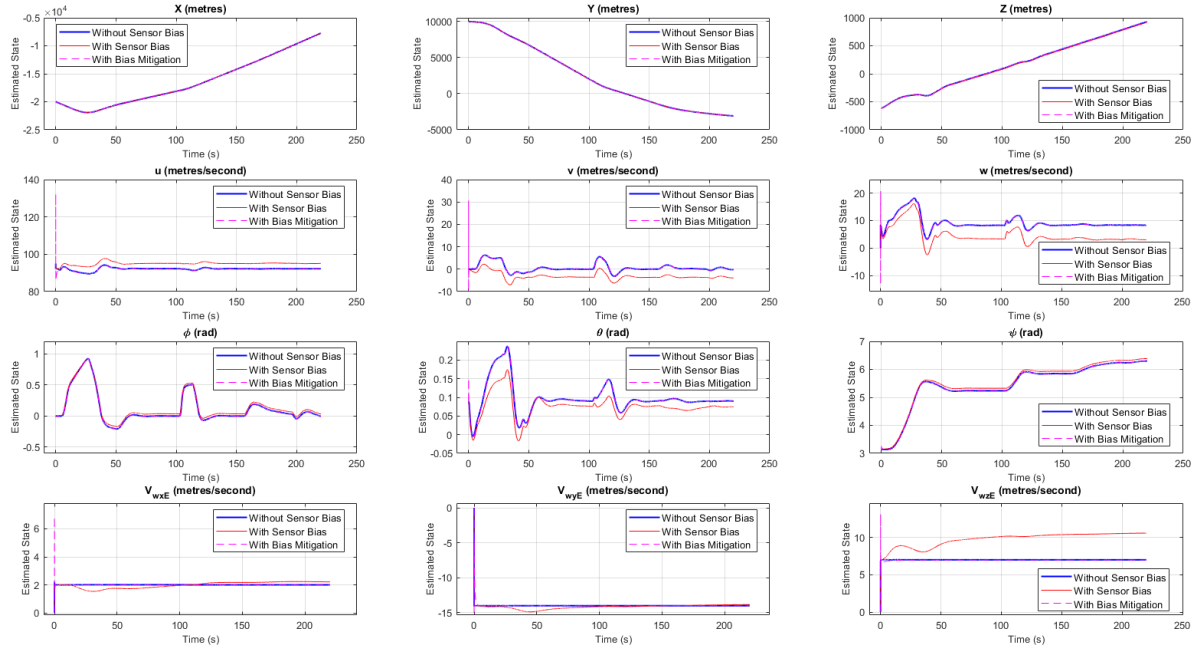


Figure 4: State Estimation after Mitigation of Bias through Kalman Filter scheme

Kalman Filter. Clear spikes are initially visible as well for some of the states indicating further that the biases are trying to be mitigated and as seen there is a clear overlapping of points in the states between the conditions when the biases have been mitigated and the sensor conditions without bias (Task1). Hence it can confidently said that the biases have been successfully mitigated.

3 Task 3: Fault Detection and Diagnose

3.1 Comparison of Sensors with and Without Fault

The use of Kalman Filter is seen in both the Task 1 and Task 2. In Task 1, it was able to use the known equations of the states of the aircraft and deliver stabilized output of the states of the aircraft through the sensors. In Task 2 the same Kalman Filter was used to mitigate the biases that occur during the reading of the GPS values from the sensors. In this task, it is shown the Kalman Filter can also be used for fault detection. Using the *dataTask3_1.mat* file a comparison can be made between the sensors without bias and faults to the one with faults to understand, how and when the fault occurs. Figure 5 shows the that first fault occurs at around the 41 seconds mark. All the states affected by the fault in the sensor shows a variation in the estimation of the values starting at this point.

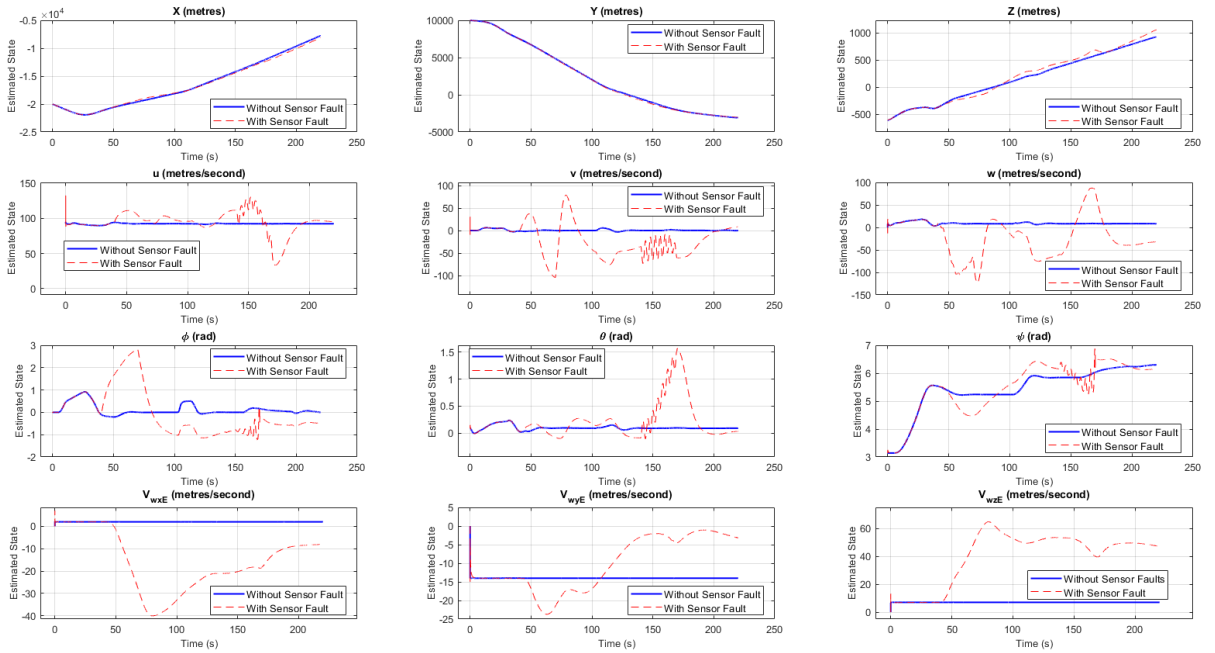


Figure 5: State Estimates of Kalman Filter Design with Sensor Fault

To have a better understanding of the fault detection by the Kalman Filter let's look at the innovation that arises from the Kalman Filter. Generally the innovation is the difference between the observed measurement (z_k) at a time k and $\hat{z}_k$ represents the predicted measurement. The following equation shows the Kalman Filter innovation:

$$v_k = z_k - \hat{z}_k \quad (4)$$

Now, looking at Figure 6, it is evident that around the 41 seconds mark the innovation is neither zero nor their mean is 0 (as shown in the case of aircraft position represented by x, y and z). The change in innovation occurs due to difference in the value provided by the sensor that occurs because of the presence of faults that occur. If there is a nominal condition of the system, the innovation is meant to be shown 0 as there is no difference between observed and predicted measurement for such a case.

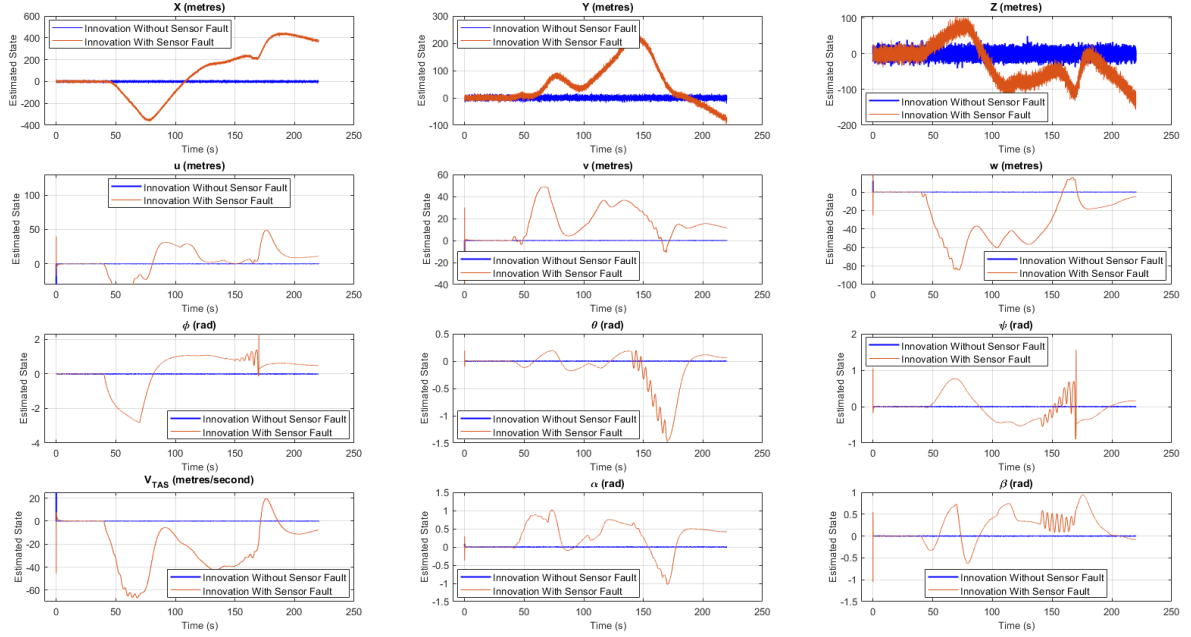


Figure 6: Innovation of Kalman Filter Design with Sensor Fault

3.2 Mitigation of Fault using Kalman Filter

Equation 4 can be used as a base for mitigating the faults that arises in the sensor. For this mitigation process 6 additional process noises were added to the 6 bias terms as shown in the equations below:

$$\omega = [\omega_{A_x}, \omega_{A_y}, \omega_{A_z}, \omega_p, \omega_q, \omega_r, \omega_{b_{A_x}}, \omega_{b_{A_y}}, \omega_{b_{A_z}}, \omega_{b_p}, \omega_{b_q}, \omega_{b_r}]$$

represents the 6 process noise that were added, and:

$$b_{A_x}(t_k) = b_{A_x}(t_k) + \omega_{b_{A_x}}(t_k), \quad b_{A_y}(t_k) = b_{A_y}(t_k) + \omega_{b_{A_y}}(t_k), \quad b_{A_z}(t_k) = b_{A_z}(t_k) + \omega_{b_{A_z}}(t_k)$$

$$b_p(t_k) = b_p(t_k) + \omega_{b_p}(t_k), \quad b_q(t_k) = b_q(t_k) + \omega_{b_q}(t_k), \quad b_r(t_k) = b_r(t_k) + \omega_{b_r}(t_k)$$

represents the bias terms after addition of the process noise. Once these process noise were dealt with after linearization and applying the Kalman Filter scheme to mitigate the faults, a similar estimation is expected for the states as shown in Figure 1. The following plots given in Figure 7 shows a better understanding of how the estimations are done using Kalman Filter for sensors when the faults are mitigated and the comparison of such a case as to when Kalman Filter is run without mitigating the faults.

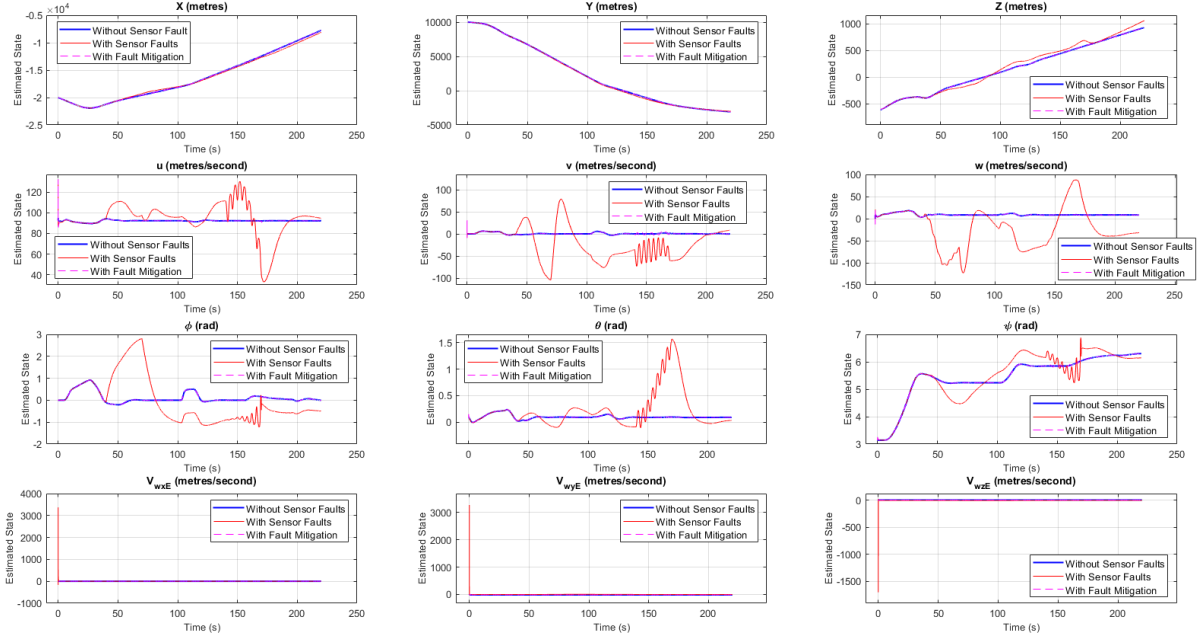


Figure 7: State Estimation after Mitigation of faults through Kalman Filter scheme

From the above figure, it is evident that after the faults have been mitigated, the estimations are the same as that of the case when running the Kalman Filter without any bias or faults as described during Task1. Now since the estimation after mitigation and the original condition of the sensor during Task1 are the same, the innovation for the former case should be around 0 or the mean should be 0. Figure 8 clearly indicates that the innovation of the Kalman Filter after mitigation of the faults in the sensor is 0 or the mean is 0 (in the case of the values represented by the values of x , y and z) similar to the case when the innovation of the Kalman Filter during Task1. The figure also shows plots for the case explained previously, when the fault is not mitigated just to understand the change that occurs in the innovation after fault mitigation.

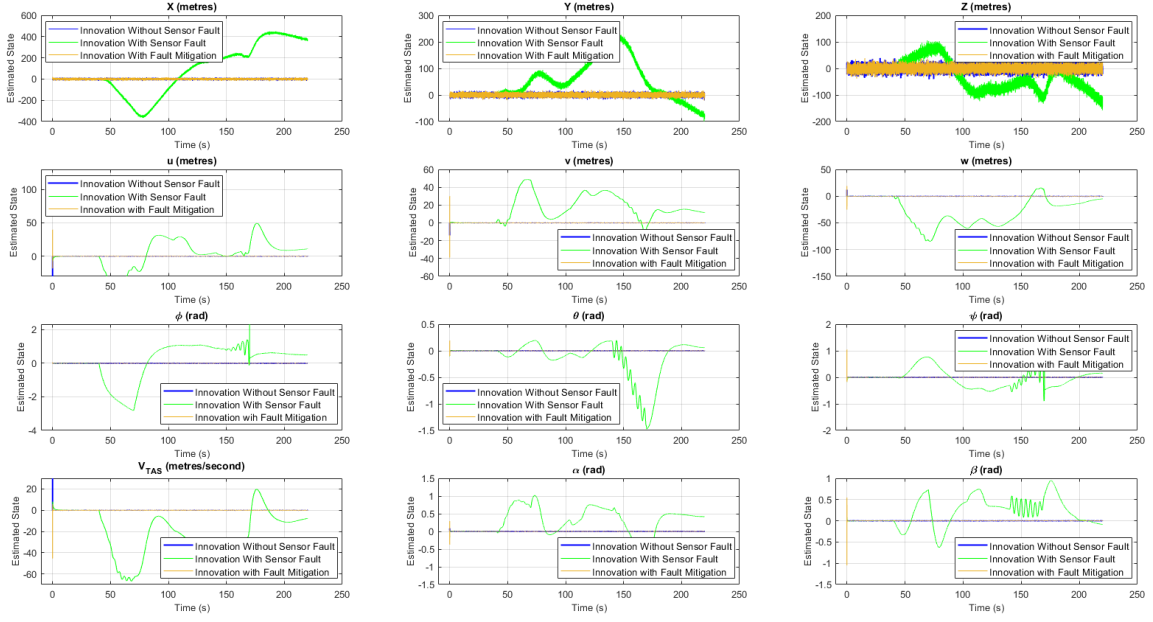


Figure 8: Innovation after Mitigation of faults through Kalman Filter scheme

3.3 Fault Detection and CUSUM Test

Both, Figure 7 and Figure 8 show that the faults have been mitigated, but the bias estimation shown in Figure 9 indicate that there are still faults that occur in the estimation of biases even though the data has been passed through an FDD based Kalman Filter scheme. Here the consideration for fault detection is considered in the cases of bias and not the inputs. This is done because as seen from the equations as discussed above, the bias has a direct relation with the measurements A_x , A_y , A_z , p , q and r . Therefore any faults that occurs affect the biases which in turn impacts the value given by the measurements. Hence, the Figure 9 shows a comparison between the faults that are detected after the FDD in the bias and the condition of the bias during an ideal case which is ideally meant to be 0.

Now to detect the faults of the measurement, a two sided CUSUM test can be used. A two sided CUSUM test is very advantageous in this condition, as it provides greater sensitivity especially when there are shifts in the model along with the minimization of false negatives. Now applying the CUSUM test on the biases of the Kalman Filter gives rise to the 6 biases faults detection as shown in the Figure 10.

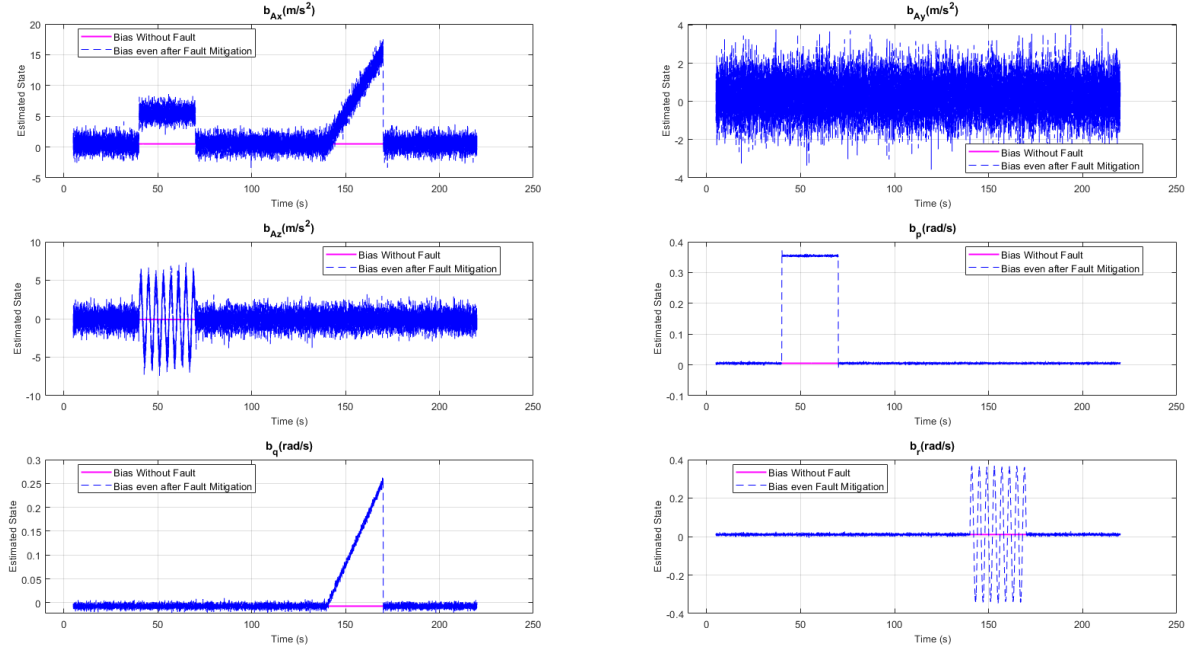
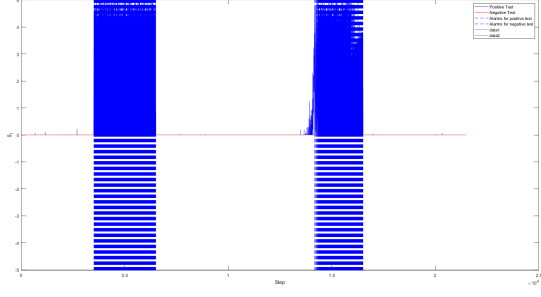
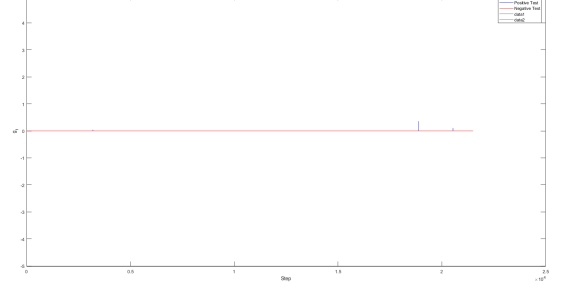


Figure 9: Faults occurring after FDD in the bias and comparison with ideal case

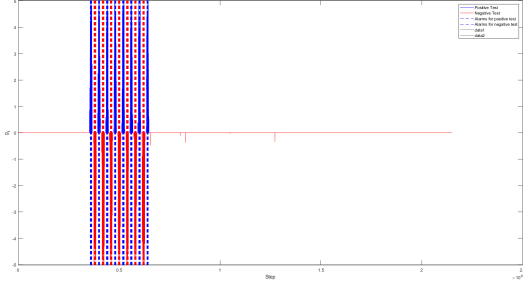
From the figure it is evident that at whatever time instance the estimation increase beyond a certain threshold, there is a fault detected. The bias estimation for b_{Ay} given by Figure 10b shows zero detection because as seen in Figure 9 there is no estimation crossing a certain threshold. The threshold for the CUSUM test for each bias is calculated by the looking at the plot in Figure 9. The problem with such a detection method is the issue false alarm rate, the detection rate and the trade-off between them. Choosing a large standard deviation resulted in the change in difference between the measured and nominal value of the measurements. On the other hand small standard deviation will effect any change in deviations but stays away from the difference of the measured and nominal value. For the test done here, a small standard deviation was considered resulting in shorter time for fault detection but leading to increase in rate of false alarm. To mitigate the initial false alarm faults, an initial 500 time instance of the data was ignored. This time skip avoids the false fault detection of the initial spike that arises from the Kalman Filter.



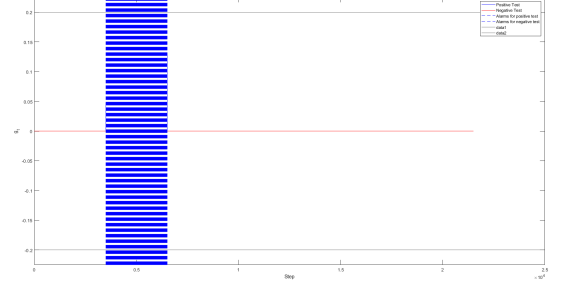
(a) A_x



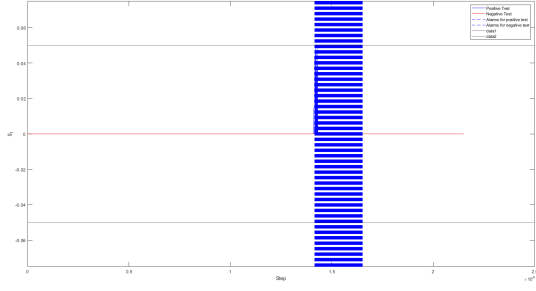
(b) A_y



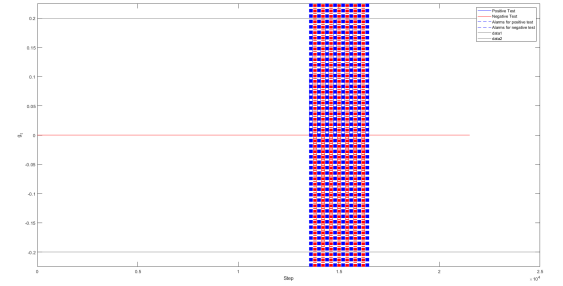
(c) A_z



(d) b_p



(e) b_q



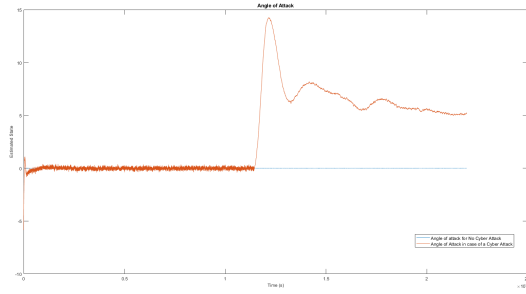
(f) b_r

Figure 10: Detection of Fault with CUSUM test

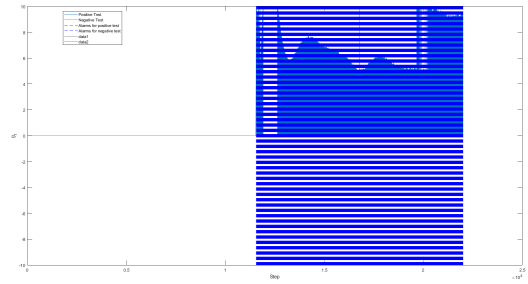
The initial fault is detected at the 4000 time instance range meaning around the 40 seconds mark (35 seconds shown in the plot, but since the initial 5 seconds time instances were ignored, we add it to this 35 seconds). This value aligns with the value that was found out during the fault detection using the innovation of the Kalman Filter as seen in the the Figure 6. Even after the mitigation through the FDD the next fault is detected at the 14000 time instance mark (again $13500 + 500$) leading to a fault detected at 140 seconds mark (These detection's are clearly seen in Figure 10).

3.4 Cyber Attack on the Angle of Attack Sensor

The dataset *dataTask3_2.mat* represents the data read by the sensors mid flight. When hit by a cyber attack during the flight time, it is seen that the values in the sensor corresponding to the angle of attack is the one that has been tampered with. Without trying to mitigate the fault in that sensor, if a Kalman Filter scheme from Task 3.3 which uses two sided CUSUM test is applied to this new dataset, the innovation of the angle of attack shows that a fault appears to be occurring after 11480 time instances(after adding the initial 30 time instance which were considered to be skipped to avoid showing and detecting the initial spike). The difference between the faulty innovation and the ideal innovation of the angle of attack is given by Figure 11a and the CUSUM test on the innovation to detect the fault is given by the Figure 11b.



(a) Comparison of innovation of Angle of Attack after said cyber attack



(b) A_y

Figure 11: CUSUM Test of the faulty Angle of Attack after cyber attack

Now that the detection of the fault was done, let's look at the process of mitigation for it. By checking the innovation for a change in the mean and standard deviation of the angle of attack (α) and defaulting the innovation to zero in case the change crosses over a certain threshold (in this case 0.15) is the FDD scheme taken for the Cyber Attack Mitigation. Figure 12 shows the angle of attack being mitigated after the cyber attack being hit versus the condition before the hit of cyber attack.

Another way to mitigate the cyber attack is to keep the standard deviation high for the angle of attack to allow the Kalman Filter to trust the scheme more than the value from the dataset hence mitigating the effect of cyber attack. But the one major drawback with this method is that, although the Kalman Filter ignores the data from the dataset, there can be a chance that the Filter can still be influenced to a certain degree from the dataset. Hence The most sure shot way to mitigate is to follow the procedure mentioned

above by considering mean and standard deviation of the innovation of angle of attack.

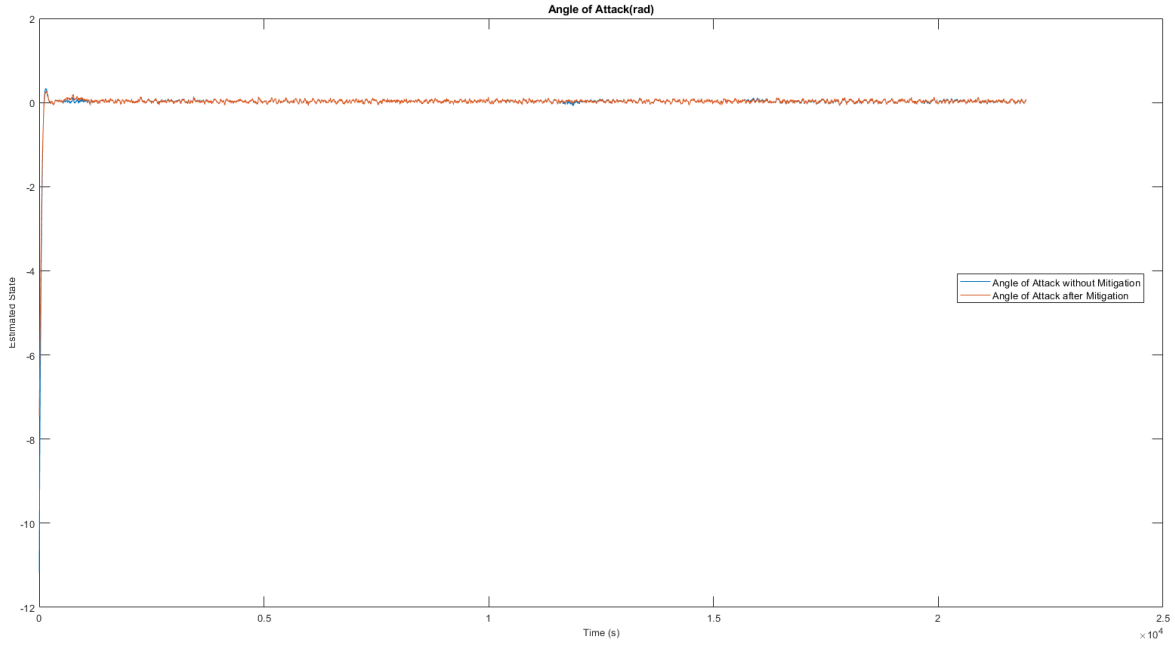


Figure 12: Angle of Attack after Cyber Attack Mitigation

From the Figure it is evident that the effect of cyber attack on α has been successfully mitigated. Now Figure 13 shows the state estimations of all the states of the system after the mitigation. The values estimated for all the states are very similar to the case in Task1 which was the basic Kalman Filter scheme without any bias and faults. Thus it can be said that the cyber attack has been successfully mitigated

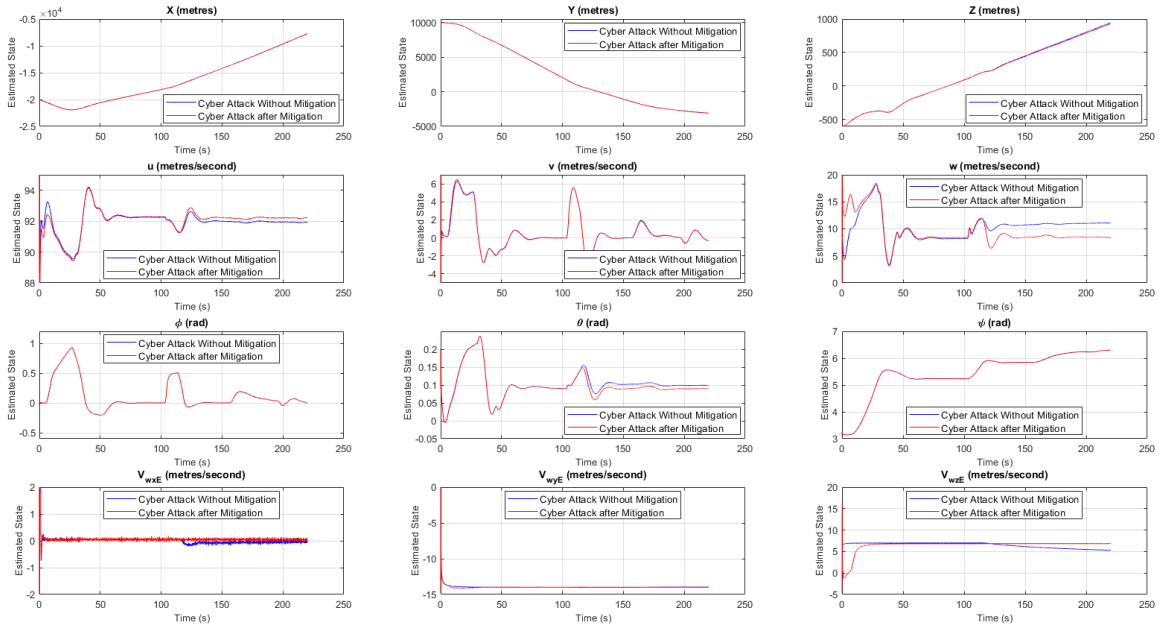


Figure 13: System States after Cyber Attack Mitigation